

Lasers and Optical fibers

Lasers: Characteristics of LASER, Interaction of radiation with matter, Expression for Energy Density and its significance. Requisites of a Laser System. Conditions for Laser action. Principle, Construction and Working of Carbon Dioxide Laser. Application of Lasers in Defense (Laser range finder) and Laser Printing. Numerical, Problems.

Optical Fibers: Total Internal Reflection, Propagation mechanism, Angle of Acceptance, Numerical Aperture, Fractional Index Change, Modes of Propagation, Number of Modes and V Number, Types of Optical Fibers. Attenuation and Mention of Expression for Attenuation coefficient, Attenuation Spectrum of an Optical Fiber with Optical Windows. Discussion of Block Diagram of Point-to-Point Communication, Intensity based Fiber Optic Displacement Sensor, Merits and Demerits, Numerical problems.

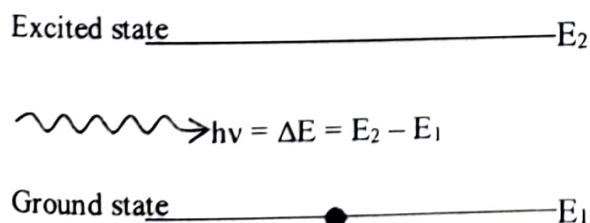
08 Hours

LASERS

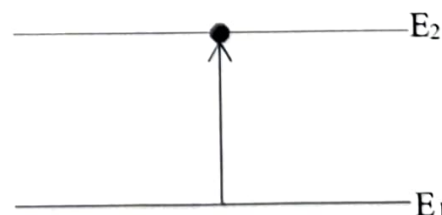
- LASER - is the acronym of Light Amplification by Stimulated Emission of Radiation.
- A laser is a device that emits light through a process of optical amplification based on stimulated emission of electromagnetic radiation.
- The first laser was built in 1960 by T. H. Maiman at Hughes research laboratory based on the theoretical work by C H Townes and A L Schawlow.
- A laser beam is highly parallel, coherent beam of light of very high intensity.
- Production of a laser light is a particular consequence of interaction of radiation with matter.
- There are three possible ways through which interaction of radiation with matter can take place and they are described as follows by considering an atom as quantum system.

(i) Induced Absorption

Induced absorption is the absorption of an incident photon by an atom. As a result of which the system is elevated from a lower energy state to higher energy state, Where the difference in energy of the two states is precisely equal to the energy of the photon.

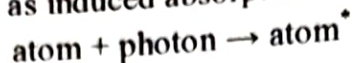


Atom in the ground state
Before the absorption of a photon



Atom in the excited state
After the absorption of a photon

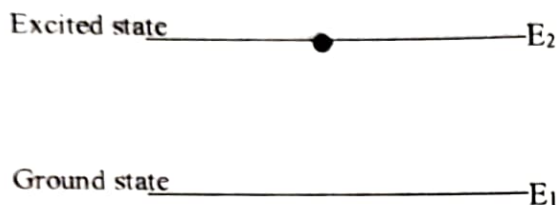
Let E_1 and E_2 be two energy levels in energy level scheme of an atom in which E_1 corresponds to a lower energy state or ground state and E_2 represents higher energy state or excited state. Let a photon of energy $\Delta E = E_2 - E_1$ incident on atom. The atom absorbs it and made a transition to the excited state. This process is called as induced absorption and is represented as



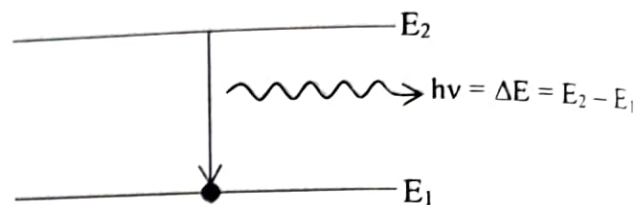
atom^* indicates that atom is in excited state.

(ii) Spontaneous emission

It is the emission of a photon when a system transits from a higher energy state to lower energy state without the aid of any external agency.

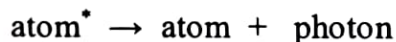


Atom in the excited state
before the emission of a photon



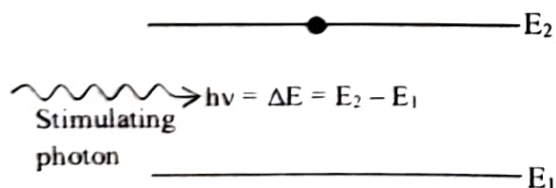
Atom emitting a photon spontaneously
by de-exciting to ground state

Consider an atom in the excited state. When it is coming back to the lowest energy state the atom voluntarily emits a photon of energy $\Delta E = E_2 - E_1$. Since the atom emits photon voluntarily without the aid of any external agent, it is called as spontaneous emission and this process is represented as,

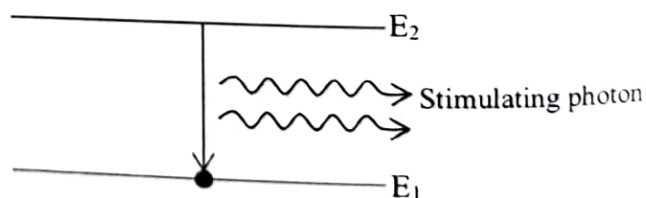


(iii) Stimulated emission

It is the emission of a photon by an excited atom by the influence of another photon, due to which the system transits from excited state to ground state. The photon thus emitted is called as stimulated photon and it will have same phase, energy and direction of movement as that of incident photon.

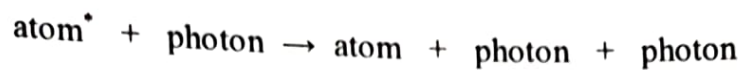


Atom in the excited state and another
photon coming near the excited atom
for stimulating the atom to emit a
photon.



Atom emitting a photon by the
influence of another photon.

Consider an atom in the excited state. Let a photon having an energy ΔE precisely equal to $E_2 - E_1$, interact with the atom by passing in its vicinity. Under such stimulation, the atom emits a photon and transits to the lower energy state. The two photons travel in exactly the same direction and have identical phase and thus they are coherent. The process can be represented as



Einstein Coefficients

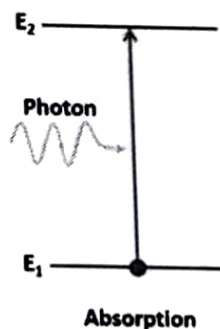
- Einstein explored the basic mechanisms involved in the interaction between radiation and matter.
- Einstein coefficients are mathematical quantities which are a measure of the probability of absorption or emission of a light by an atom or molecule.
- The Einstein 'A' coefficient is related to the rate of spontaneous emission of light and 'B' coefficients are related to the absorption and stimulated emission of light.

Expression for energy density of radiation under equilibrium condition in terms of Einstein coefficients

- Consider two energy states E_1 and E_2 of a system of atoms such that ($E_2 > E_1$).
- N_1 and N_2 be the number of atoms with energy E_1 and E_2 per unit volume of the system which are called as number density of atoms in the state 1 and 2 respectively.
- Let the radiations with continuous spectrum of frequencies be incident upon the system. U_ν be the energy density of radiations of frequency ' ν '.
- Then $U_\nu d\nu$ will be the energy density of radiations whose frequencies lie in the range ν and $\nu + d\nu$.

Induced absorption

In this process an atom in the ground state can go to the higher energy state by absorbing a radiation of frequency ' ν ' such that $\nu = \frac{E_2 - E_1}{h}$,



The number of such absorptions per unit volume is called "rate of absorption". The rate of absorption depends upon the number density of lower energy state N_1 and the energy density U_ν .

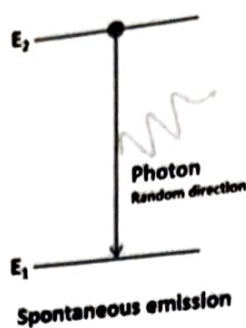
$$\therefore \text{Rate of absorption} \propto N_1 U_\nu$$

$$\text{Rate of absorption} = B_{12} N_1 U_\nu \quad (1)$$

where, $B_{12} \rightarrow$ Einstein coefficient of induced absorption

Spontaneous emission

In this process an atom in the higher energy state undergoes a transition to the lower energy state voluntarily by emitting a photon.



Since it is a voluntary transition, it is independent of the energy density of any frequency in the incident radiation. The number of such spontaneous emissions per unit time per unit volume is called "rate of spontaneous emission" which is proportional to only the number density in the higher energy state, i.e. N_2 .

$$\therefore \text{Rate of spontaneous emission} \propto N_2$$

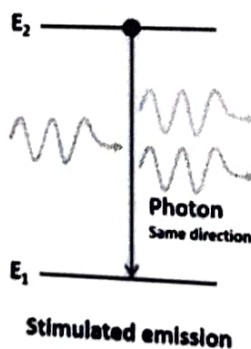
$$\text{Rate of spontaneous emission} = A_{21}N_2 \quad (2)$$

where, $A_{21} \rightarrow$ Einstein coefficient of spontaneous emission

Stimulated emission

In this process the system requires an external photon of suitable frequency $\nu = \frac{E_2 - E_1}{h}$

to stimulate the atom for the corresponding downward transition and thereby cause emission of stimulated photon.



The number of stimulated emissions per unit time per unit volume is called rate of stimulated emission.

$$\therefore \text{Rate of stimulated emission} \propto N_2 U_\nu$$

$$\text{Rate of stimulated emission} = B_{21}N_2 U_\nu$$

where, $B_{21} \rightarrow$ Einstein coefficient of stimulated emission.

(3)

At thermal equilibrium,

$$\text{Rate of absorption} = (\text{Rate of spontaneous emission} + \text{Rate of stimulated emission})$$

$$B_{12}N_1 U_\nu = A_{21}N_2 + B_{21}N_2 U_\nu$$

$$U_\nu (B_{12}N_1 - B_{21}N_2) = A_{21}N_2$$

$$U_\nu = \frac{A_{21}N_2}{(B_{12}N_1 - B_{21}N_2)}$$

$$\text{i.e. } U_\nu = \frac{A_{21}}{B_{21}} \left[\frac{N_2}{\left(\frac{B_{12}}{B_{21}} N_1 - N_2 \right)} \right]$$

$$U_\nu = \frac{A_{21}}{B_{21}} \left[\frac{1}{\left(\frac{B_{12}N_1}{B_{21}N_2} \right) - 1} \right] \rightarrow (4)$$

By Boltzmann's law, $N_2 = N_1 e^{-\left(\frac{E_2 - E_1}{KT}\right)} = N_1 e^{-h\nu/KT}$

$$\text{i.e., } N_1/N_2 = e^{h\nu/KT}$$

Eqn. (4) becomes
$$U_\nu = \frac{A_{21}}{B_{21}} \left[\frac{1}{\left(\frac{B_{12}}{B_{21}} e^{\left(\frac{h\nu}{KT}\right)} - 1 \right)} \right] \rightarrow (5)$$

By Planck's law,
$$U_\nu = \frac{8\pi h \nu^3}{c^3} \left[\frac{1}{\left(e^{\left(\frac{h\nu}{KT}\right)} - 1 \right)} \right] \rightarrow (6)$$

Comparing equation (5) & (6)

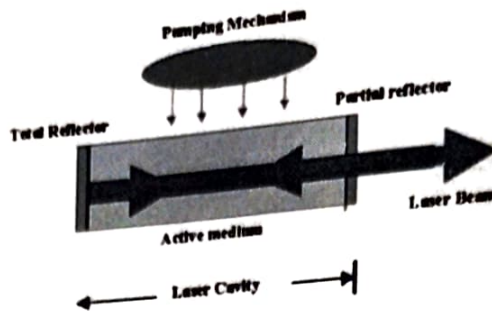
$$\frac{A_{21}}{B_{21}} = 8\pi h \nu^3 / c^3 \quad \& \quad \frac{B_{12}}{B_{21}} = 1 \quad \text{i.e. } B_{12} = B_{21}$$

The probability of induced absorption is equal to the stimulated emission.
Therefore A_{12} is written as A and B_{12} , B_{21} written as B .

Equation (5) becomes
$$U_\nu = \frac{A}{B} \left[\frac{1}{\left(e^{\left(\frac{h\nu}{KT}\right)} - 1 \right)} \right]$$

This is the expression for energy density of radiation in terms of Einstein coefficients.

Requisites of a Laser System



There are three important requisites of a laser system and they are as follows,

(i) Laser cavity

A laser cavity is a closed cavity which is filled with an active medium along with two mirrors. It provides the feedback necessary to tap certain amount of laser energy from the active medium. The feedback action is achieved through the mirrors attached to the cavity. The photons undergo multiple reflections from the mirrors and during this time the photons stimulate more and more atoms to emit photons with identical nature.

In order to avoid destructive interference of the photon of the laser beam, the distance 'L' between the mirrors should be such that the cavity should support an integral number of half wave length i.e., $L = n(\lambda/2)$.

(ii) Active medium

An active medium is one which consists of atoms or molecules which are having metastable states and thereby supports for population inversion, which is necessary for laser emission. Stimulated emission from the atoms of the active medium gives the laser light.

(iii) Excitation source

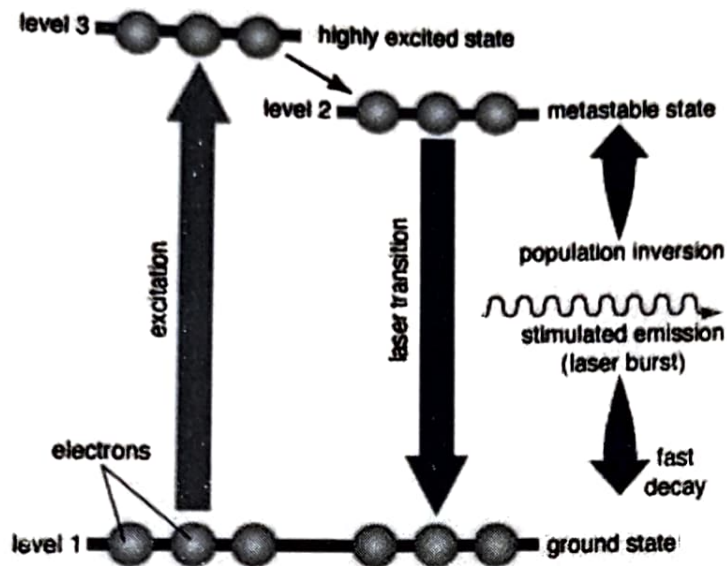
The excitation source provides the appropriate amount of energy for pumping the atoms of an active medium to the higher energy levels. The input energy may be in the form of light energy, which is known as optical pumping and is made use in the construction of ruby laser. The pumping may be achieved by electrical energy input in which is electrical pumping. In He-Ne laser the pumping is achieved by electrical discharge.

Condition for laser action

Population inversion and metastable state

- Population inversion is the state of a system at which the population of a particular higher energy state is more than that of the specified lower energy state.
- In real physical system, the population inversion conditions do not exist under normal conditions. However, it is possible to achieve the population inversion condition in certain systems which possess a special kind of excited states called "metastable states".
- If an atom is made to excite for a higher energy state, the atom returns into the ground state within a short time not more than 10^{-8} second.

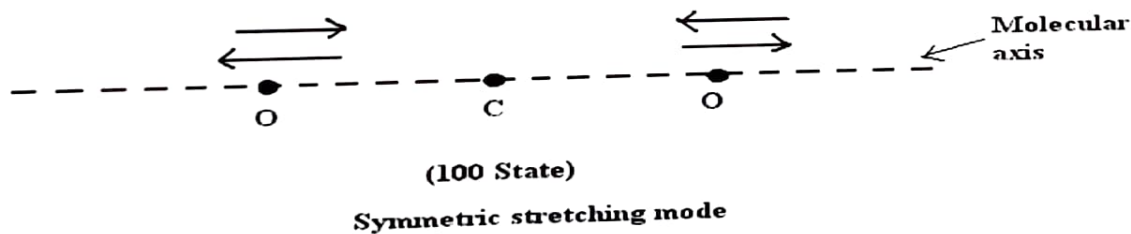
- If the excited state is a metastable state then the atom stays for an unusually long time, which is of the order of 10^{-3} - 10^{-2} second.
- Therefore we can have more number of atoms in the excited state than the ground state. Thus we can achieve population inversion.



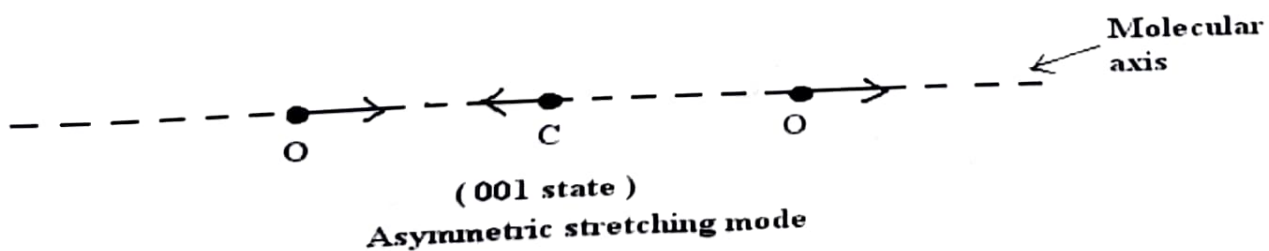
Vibrational modes of CO₂ molecule:

A carbon dioxide molecule has two oxygen atoms between which there is a carbon atom. It has 3 different modes of vibration.

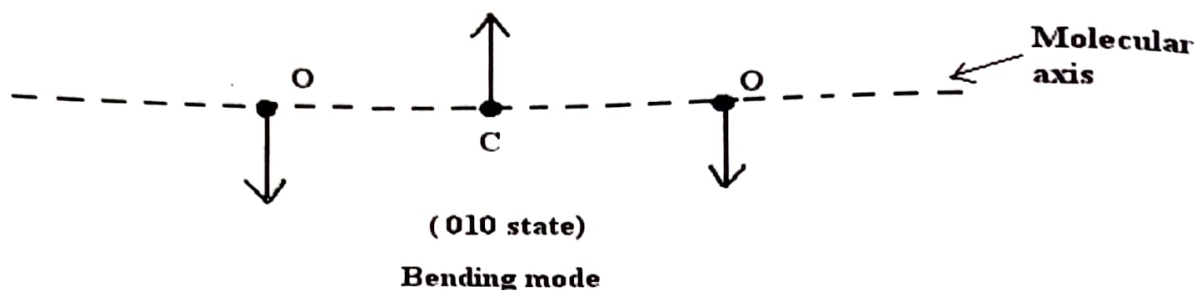
1. **Symmetric stretching mode** : In this mode, carbon atom is stationary and the oxygen atoms oscillate to and fro along the molecular axis. This state is referred as (100) state.



2. **Asymmetric stretching mode**: In this mode, both the oxygen atoms move in one direction while the carbon atom moves in opposite direction along the molecular axis. This state is referred as (001) state.



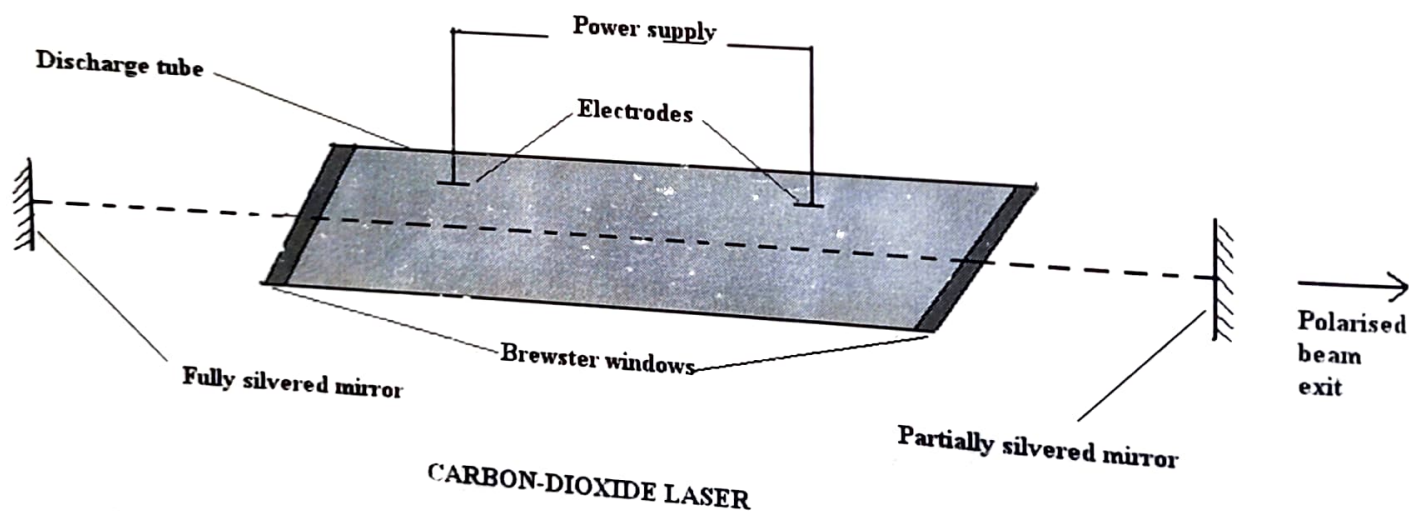
3. **Bending mode:** In this mode, the carbon atom and oxygen atoms moves perpendicular to molecular axis in the opposite direction. This state is referred to as (010) state.



CO₂ LASER:

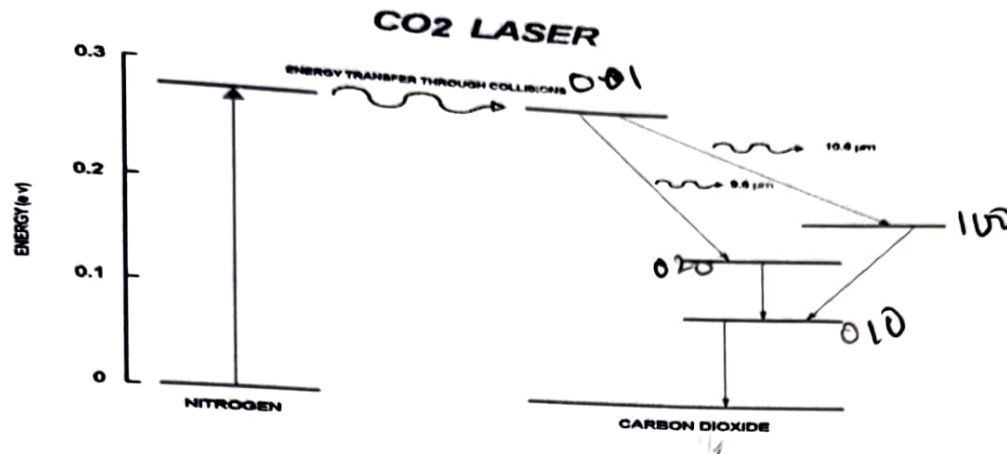
It was devised by C.K.N Patel in 1924. CO₂ laser is molecular gas laser which operates in the IR region involving a set of vibrational – rotational transitions. It is a four level laser producing both continuous and pulsed laser.

Construction:



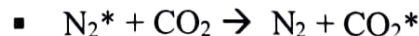
- It consists of discharge tube (quartz) of diameter 2.5cm and length of 5m.
- The tube is filled with a mixture of CO₂, N₂ and He gas in the ratio 1:2:3.
- Sometimes water vapour is added because during discharge CO₂ molecule breaks up into CO and O. The water vapour additives help in deoxidizing CO to CO₂.
- Brewster window made up of flat quartz are sealed to the tube at both of its ends to give polarized light.
- The tube has got two parallel mirrors. One is partially silvered and the other is fully silvered to function as laser cavity

Working:



CO₂ laser Energy level diagram

- When An electric field is applied electrons rendered free from atoms, collide with N₂ & CO₂ molecules in their path towards positive electrodes due to which N₂ atoms are excited to the higher energy level v_1 .
- Likewise it happens to the CO₂ molecule. This is collision of first kind
 - $e_1 + N_2 \rightarrow e_2 + N_2^*$
 - $e_1 + CO_2 \rightarrow e_2 + CO_2^*$
- Where, e_1 and e_2 are the energies of electrons before and after collisions.
- Let the ground state, (010) state, (020) state, (100) state and (001) are represented as E_1, E_2, E_3, E_4 and E_5 levels respectively
- Because of matching energy levels, $v = 1$ state of N₂ is equal to (001) state of CO₂, N₂ molecule in the metastable state collide with the CO₂ in the ground state and transfer of energy takes place from N₂ to CO₂. As a result of which CO₂ molecule moved to (001) state where as the N₂ molecule moved to ground state. This is the collision of second kind.

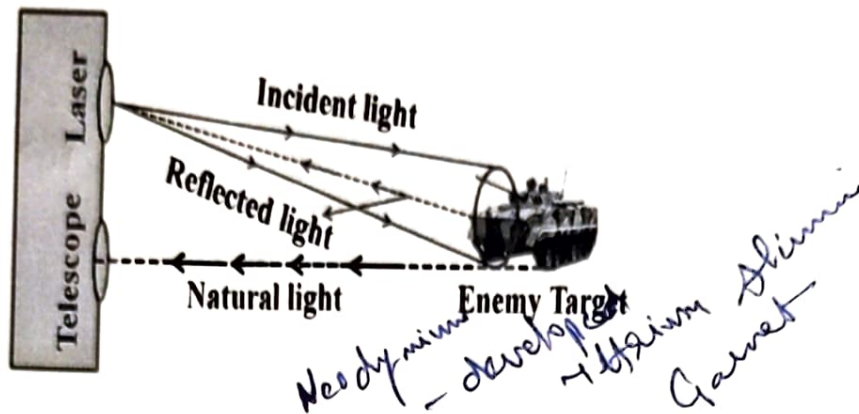


Where, CO_2 and CO_2^* are the energies of CO₂ in ground state and excited states.

- Because of the resonant transfer of energy, the population inversion is achieved in (001) state with respect to (100) and (020)
- The transition from E_5 to E_4 levels gives wavelength of 10.6 μm (in far IR region)
- The transitions from E_5 to E_3 level gives wavelength of 9.6 μm (in far IR region)
- Following these transitions the CO₂ molecules in E_4 and E_3 collide with the ground state CO₂ molecules (because of the matching energy levels) and arrive at E_2 state.
- The molecules in the E_2 state collide with He and water vapour molecules, so that come down to the ground state.
- The cycle of operation gives both continuous and pulsed laser.

Application of lasers

1. Laser in defense (Laser Range Finder)

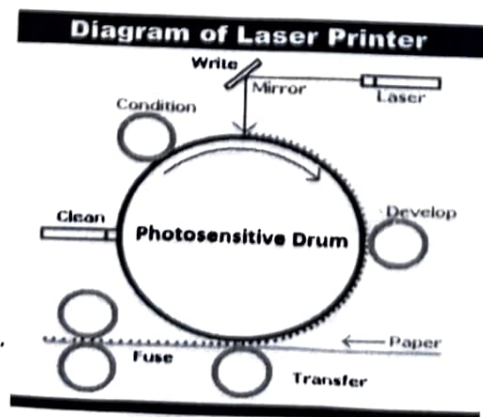


- A high-power pulsed laser (Nd-YAG) beam is directed towards the enemy target from the transmitter. The beam bounces back from the surface of the target as a reflection. A part of the reflected beam is received as a signal by the receiver. The unwanted noise signal will be filtered by the optical filter and pure signal is amplified by the photomultiplier in the receiver. The range finders high speed clock measures the exact time of incident and reflection of the pulse and then convert it in to distance.
- This will acquire only position of enemy without personal details. The laser range-finders are best hand-held equipment for modern infantry in defence sector.
- In recent years all military battle tankers are equipped with laser range-finder interface with computer for exact information of enemy in a digital readout form.
- The distance obtained from this range-finder will have less than 1% error as compare to actual distance of enemy. This can also be used for continuous tracking and ranging of missiles as well as aircrafts from ground or air.

2. Laser Printing

Laser printers were invented at XEROX in 1969 by researcher Gary Starkweather. Laser Printers are digital printing devices that are used to create high quality text and graphics on plain paper. A diode laser is used in the process of printing in LASER Printer.

Construction



Working:

1. A laser beam projects an image of the page to be printed onto an electrically charged rotating photosensitive drum coated with selenium. *(which has good photo conductivity properly spread on drum. then electric charge applied to selenium acts as photocopy)*
2. Photo conductivity allows charge to leak away from the areas which are exposed to light and the area gets positively charged.
3. Toner particles are then electrostatically picked up by the drum's charged areas, which have been exposed to light.
4. The drum then prints the image onto paper by direct contact and heat, which fuses the link to the paper.

Advantages

1. Laser printers are generally quiet and fast.
2. Laser printers can produce high quality output on ordinary papers.
3. The cost per page toner cartridges is lower than other printers.

Disadvantages

1. The initial cost of laser printers can be high.
2. Laser printers are more expensive than dot-matrix printers and ink-jet printers.

↓
toner particles
helps the
photocopy
to form
the text
and images

Working also the exposure
the Laser is activated and